# MapReduce Pipeline Documentation

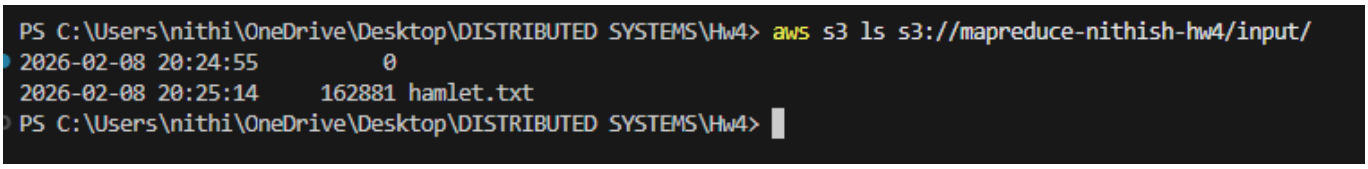
## Step 1: Input Data Upload to S3

We downloaded the Hamlet text file and uploaded it to an S3 bucket under the `input/` prefix.

**S3 Input Path**

```
s3://mapreduce-nithish-hw4/input/hamlet.txt
```

This file served as the input to the splitter task.

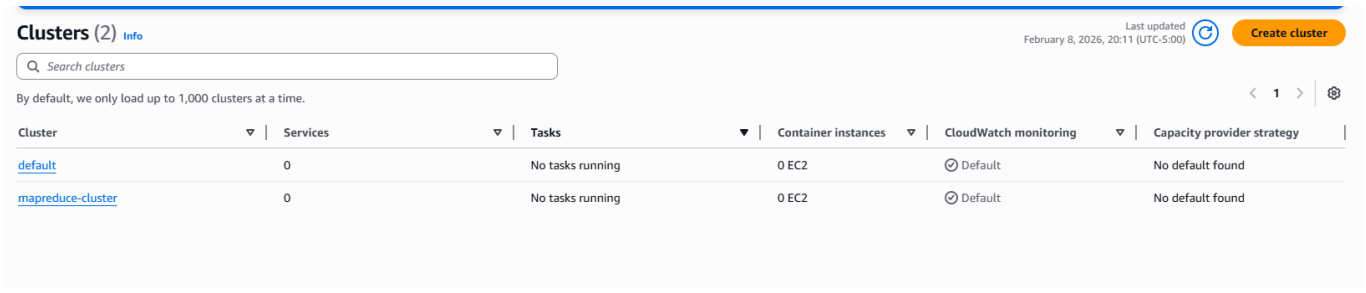



- S3 bucket showing `input/hamlet.txt`

## Step 2: ECS Cluster Setup

We created an ECS cluster named `mapreduce-cluster` using the Fargate launch type, allowing us to run containers without managing EC2 instances.

The cluster was used to run all tasks in the pipeline.




- ECS clusters page showing `mapreduce-cluster`

## Step 3: Splitter Task Execution

The splitter task was invoked via HTTP with the S3 input file URL and a chunk count of 3.

**Request**

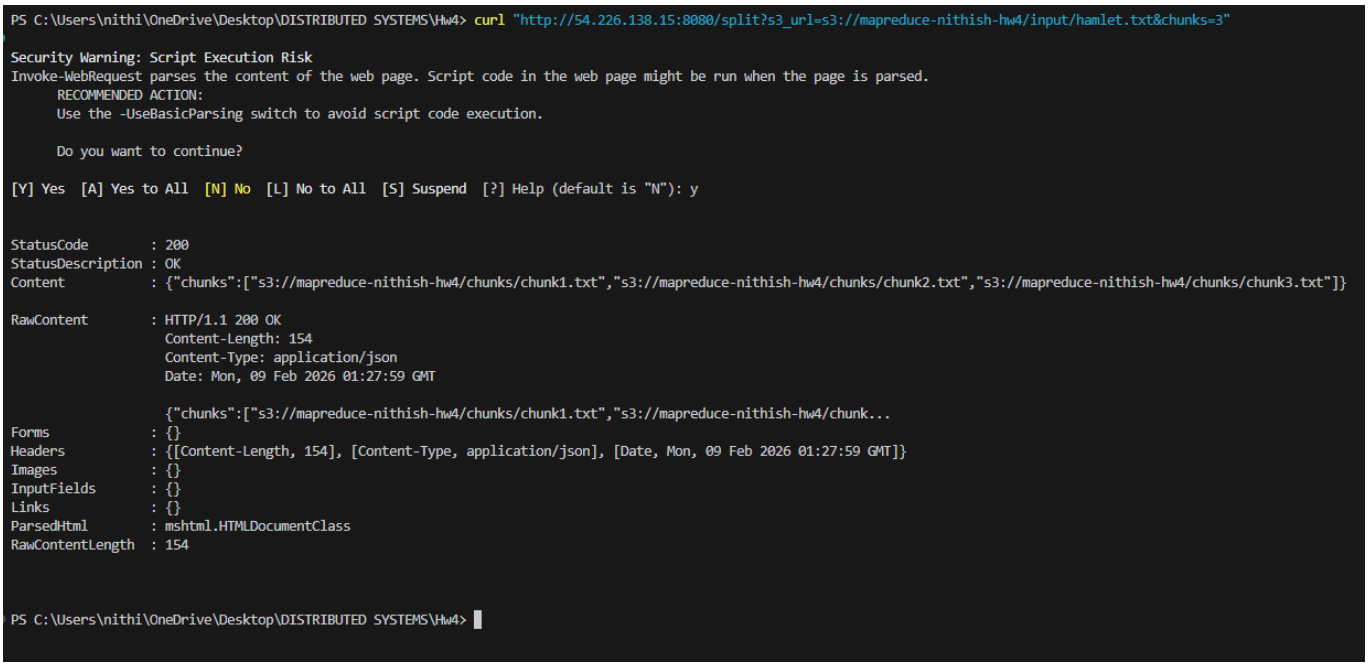
```
http://<splitter-ip>:8080/split?s3_url=s3://mapreduce-nithish-
hw4/input/hamlet.txt&chunks=3
```

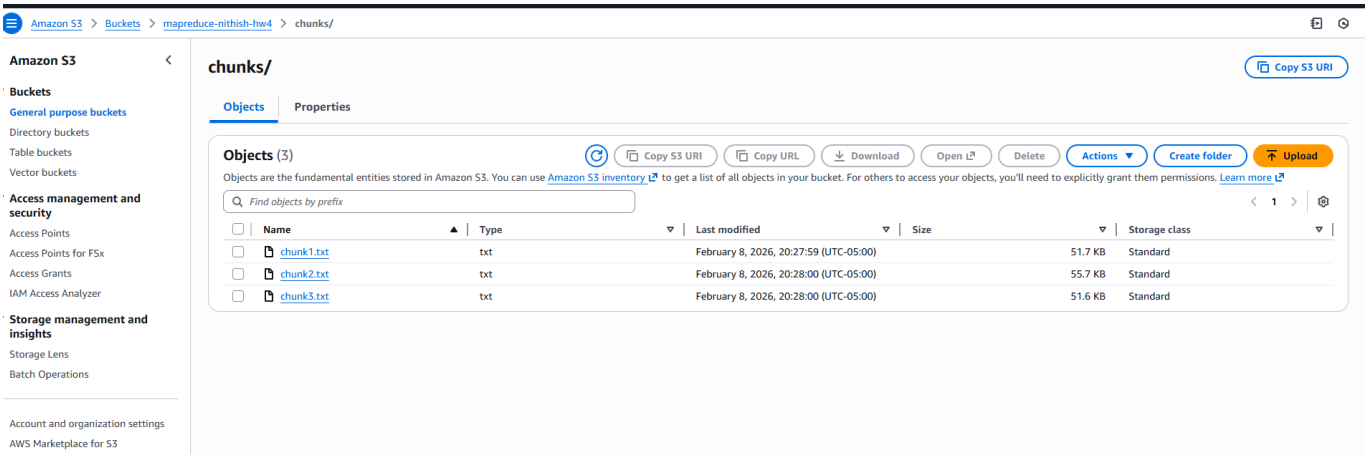
**Response**

```
{
  "chunks": [
    "s3://mapreduce-nithish-hw4/chunks/chunk1.txt",
    "s3://mapreduce-nithish-hw4/chunks/chunk2.txt",
    "s3://mapreduce-nithish-hw4/chunks/chunk3.txt"
  ]
}
```

The splitter successfully:

- Read the input file from S3
- Split it into three roughly equal-sized chunks
- Uploaded each chunk back to S3




- Curl output showing splitter response
- S3 `chunks/` folder with `chunk1.txt`, `chunk2.txt`, `chunk3.txt`

## Step 4: Mapper Task Execution

Three mapper tasks were executed independently, each processing one chunk.

Each mapper:

- Read its assigned chunk from S3
- Counted word occurrences
- Wrote results as a JSON file to S3

**Example Mapper Request**

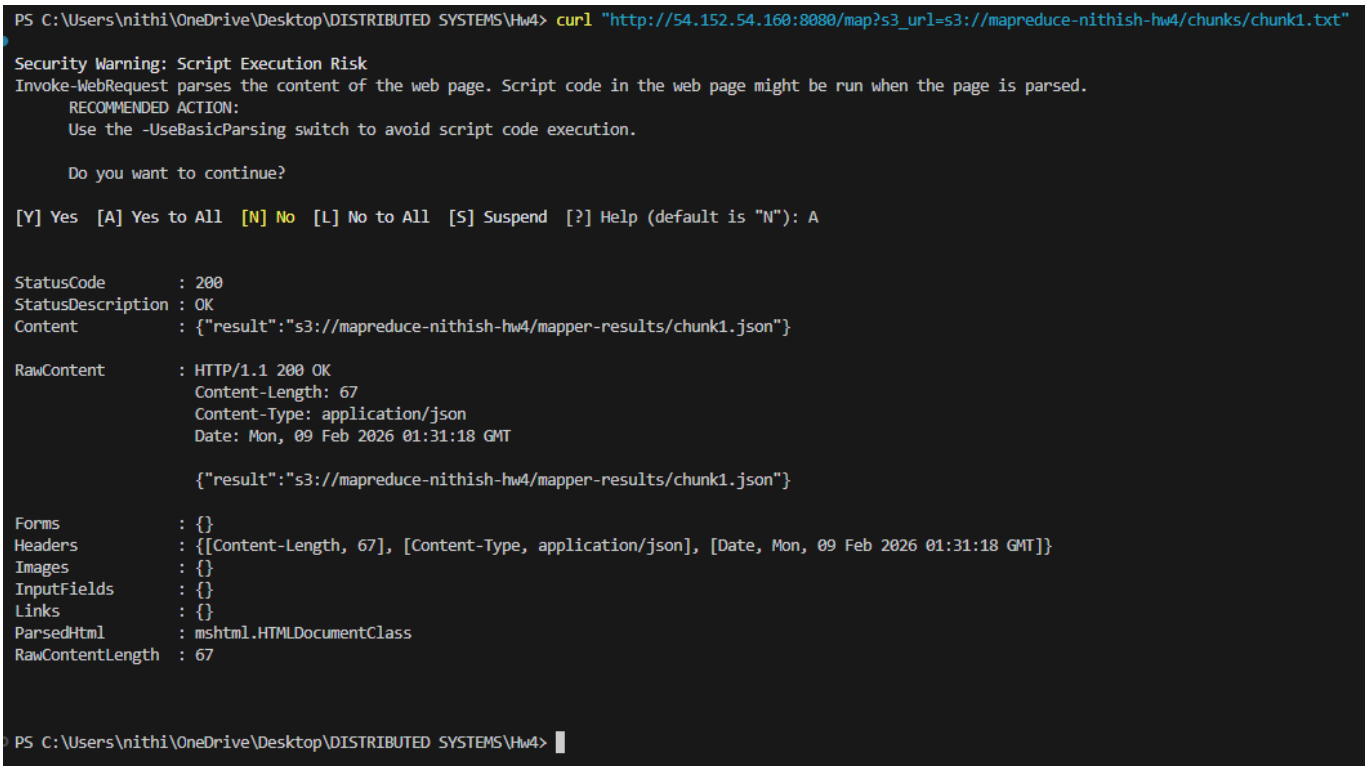
```
http://<mapper-ip>:8080/map?s3_url=s3://mapreduce-nithish-hw4/chunks/chunk1.txt
```

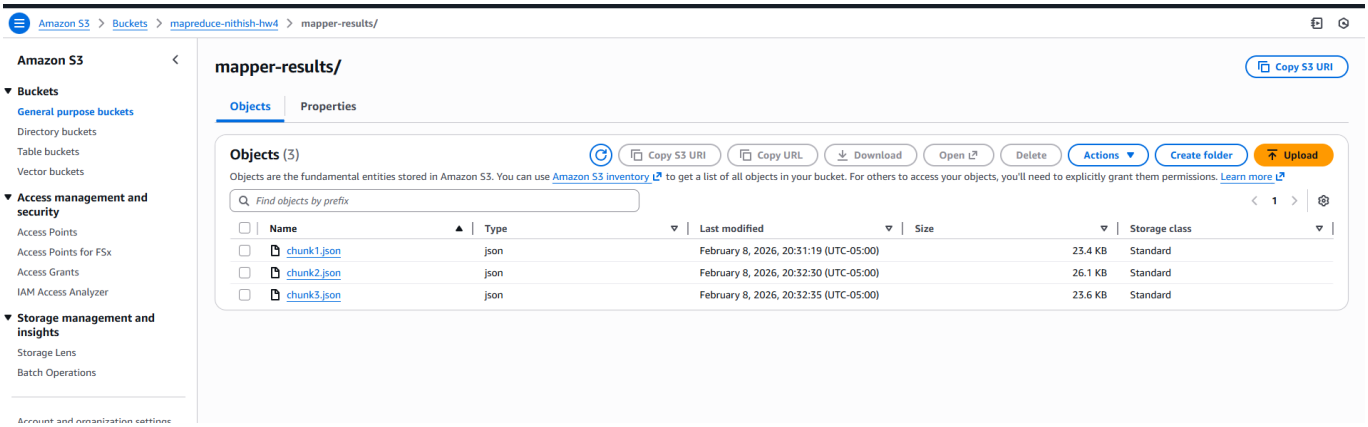
**Mapper Output**

```
{
  "result": "s3://mapreduce-nithish-hw4/mapper-results/chunk1.json"
}
```

After running all three mappers, the following files were produced:

```
s3://mapreduce-nithish-hw4/mapper-results/chunk1.json
s3://mapreduce-nithish-hw4/mapper-results/chunk2.json
s3://mapreduce-nithish-hw4/mapper-results/chunk3.json
```




- Curl output for mapper invocation
- S3 `mapper-results/` folder showing all three JSON files
- CLI `aws s3 ls` output verifying mapper results

# Step 5: Reducer Task Execution

The reducer task was invoked with the three mapper result URLs as query parameters.

### Request

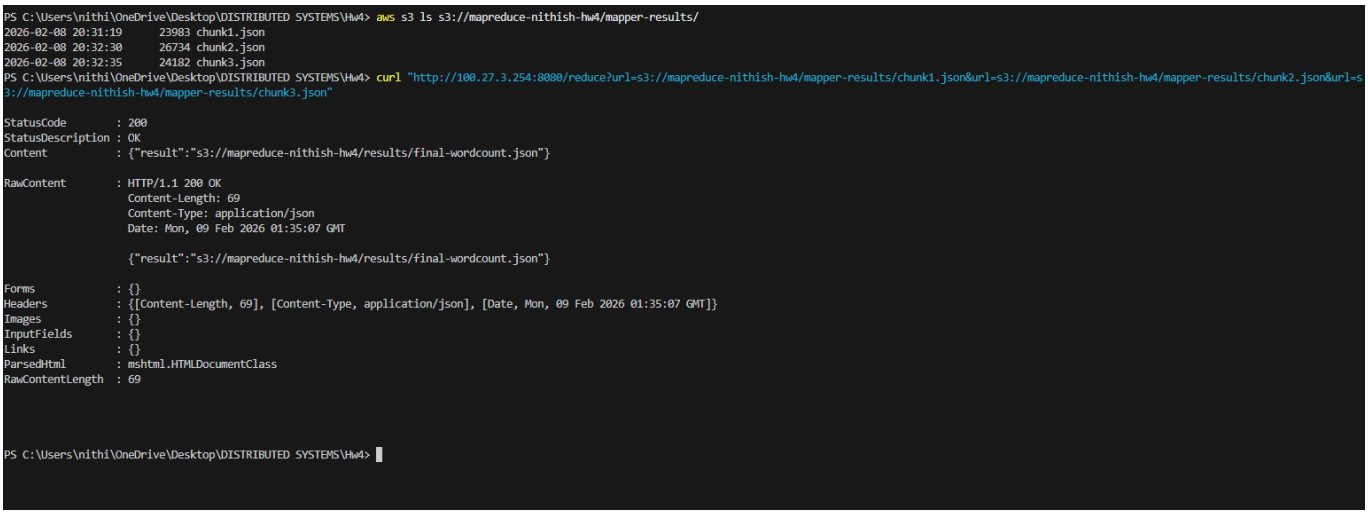
```
http://<reducer-ip>:8080/reduce?url=s3://mapreduce-nithish-hw4/mapper-
results/chunk1.json
&url=s3://mapreduce-nithish-hw4/mapper-results/chunk2.json
&url=s3://mapreduce-nithish-hw4/mapper-results/chunk3.json
```

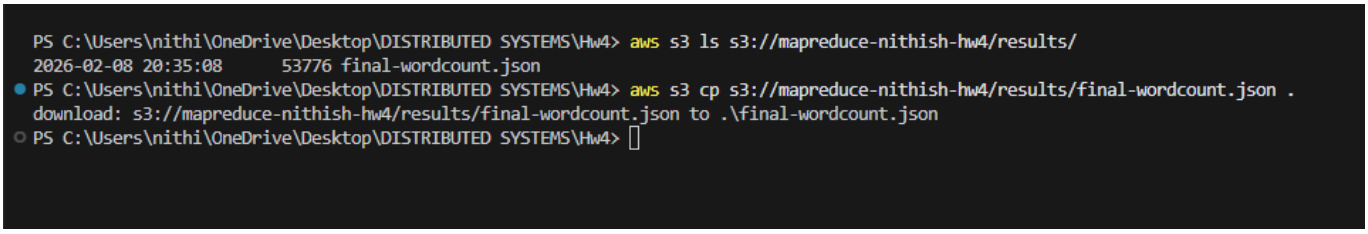
### Response

```
{
  "result": "s3://mapreduce-nithish-hw4/results/final-wordcount.json"
}
```

The reducer:

- Read all mapper outputs from S3
- Aggregated word counts
- Wrote the final combined result back to S3




- Curl output for reducer request
- S3 `results/` folder showing `final-wordcount.json`

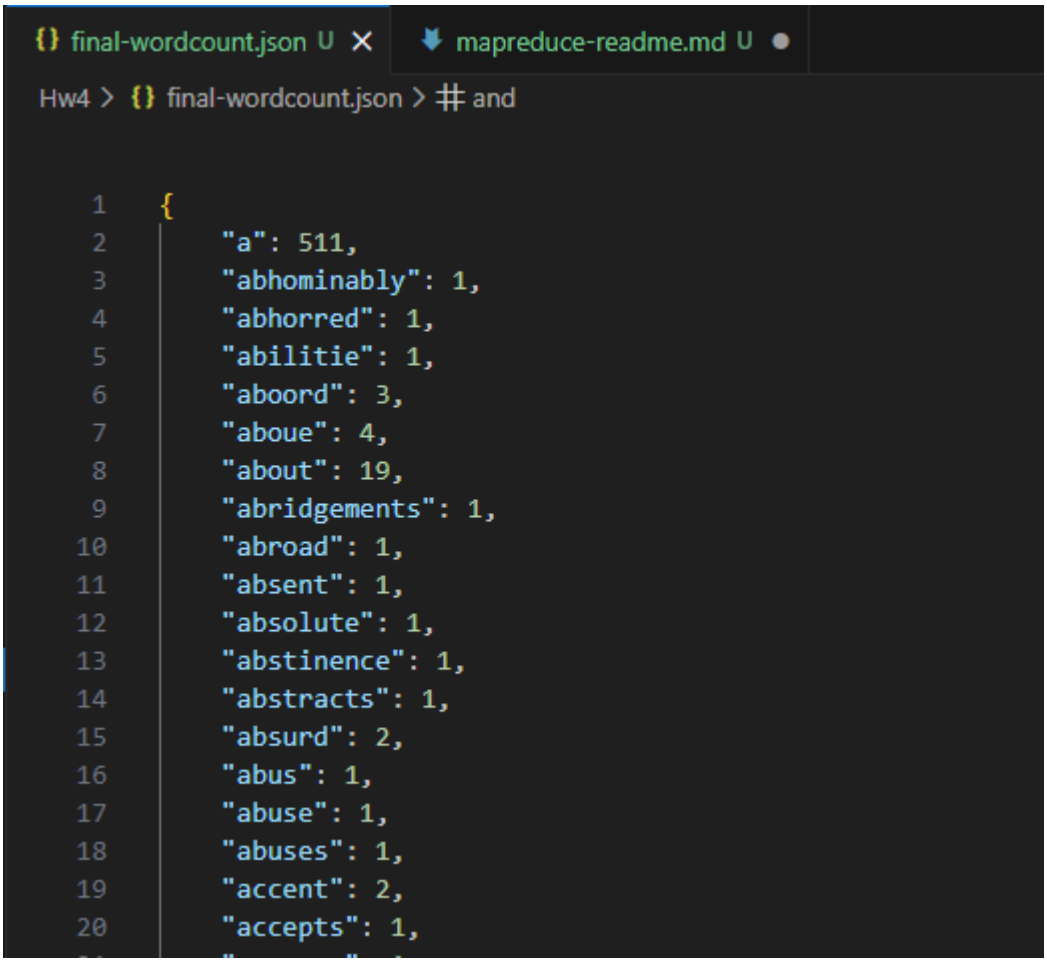
## Step 6: Result Verification

The final result file was downloaded locally and formatted using VS Code for readability.

**Final Output Location**

```
s3://mapreduce-nithish-hw4/results/final-wordcount.json
```

The JSON correctly reflects aggregated word counts across all chunks, validating the correctness of the MapReduce pipeline.




- Beautified `final-wordcount.json` opened in VS Code

## Observations and Learnings

- The workload was embarrassingly parallel, making it ideal for MapReduce.
- ECS Fargate made scaling trivial by running multiple mappers independently.
- S3 acted as a reliable shared storage medium for coordination.
- Manual orchestration highlighted the complexity of scheduling and fault handling in distributed systems.